

Cluster 0

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Subjects

Research fields: Materials science, Engineering, Chemistry

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Chemical engineering, Optoelectronics, Analytical chemistry

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: Perovskite solar cells, solar cells, inverted perovskite solar, power conversion efficiency, efficient inverted perovskite

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

0.17% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.01% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

9751 recent articles in the cluster

9751 articles in this cluster were published in the five year period ending in 2023.

Average article is 5.39 years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: 62.16th percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 62.16% of other clusters.

Citation rating: 71.89th percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 71.89% of all other articles published in the same year.

96.67% of articles are in English

Expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 9447 of 9751 articles. 8146 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	5294	6.48
United States	971	9.36
South Korea	843	10.22
India	597	3.63
Germany	475	8.62
United Kingdom	404	7.84
Japan	382	6.01
Taiwan	298	4.28
Saudi Arabia	277	6.23
Switzerland	271	14.68

Top funding countries

Data available for 7565 of 9751 articles. 6733 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	4604	6.92
South Korea	668	7.35
United States	583	10.19
Germany	287	9.26
Belgium	280	6.50
India	236	4.13
Spain	216	5.93
United Kingdom	211	9.15
Taiwan	199	3.84
Japan	186	5.73

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Managing grains and interfaces via ligand anchoring enables 22.3%-efficiency inverted perovskite solar cells](#)

2020: Nature Energy. 1011 citations.

[Perovskite solar cells with atomically coherent interlayers on SnO₂ electrodes](#)

2021: Nature. 2540 citations.

[Conformal quantum dot-SnO₂ layers as electron transporters for efficient perovskite solar cells, 2022](#)

2022: N.Y.). 1049 citations.

[Efficient perovskite solar cells via improved carrier management](#)

2021: Nature. 2342 citations.

[Controlled growth of perovskite layers with volatile alkylammonium chlorides](#)

2023: Nature. 1172 citations.

[Pseudo-halide anion engineering for \$\alpha\$ -FAPbI₃ perovskite solar cells](#)

2021: Nature. 2463 citations.

[Inactive \(PbI₂\)₂RbCl stabilizes perovskite films for efficient solar cells, 2022](#)

2022: N.Y.). 942 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[The Progress of Additive Engineering for CH₃NH₃PbI₃ Photo-Active Layer in the Context of Perovskite Solar Cells](#)

2021: Crystals. 26 citations.

[Advances of Electron Transport Materials in Perovskite Solar Cells: Synthesis and Application](#)

2020: 化学进展. 8 citations.

[Recent Progress in Coating Methods for Large-Area Perovskite Solar Module Fabrication](#)

2023: Solar RRL. 8 citations.

[Advancement in Inorganic Hole Transport Materials for Inverted Perovskite Solar Cells](#)

2020: Journal of Electronic Materials. 37 citations.

[Recent Advances on Printing Technology for Perovskite Photovoltaics](#)

2023: 硅酸盐学报. 0 citations.

[Progress toward understanding the fullerene-related chemical interactions in perovskite solar cells](#)

2022: 纳米研究. 16 citations.

[Recent progress in electron transport bilayer for efficient and low-cost perovskite solar cells: a review](#)

2021: Journal of Solid State Electrochemistry. 19 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Managing grains and interfaces via ligand anchoring enables 22.3%-efficiency inverted perovskite solar cells](#)

2020: Nature Energy. 1011 citations.

[Perovskite solar cells with atomically coherent interlayers on SnO₂ electrodes](#)

2021: Nature. 2540 citations.

[Impact of strain relaxation on performance of alpha-formamidinium lead iodide perovskite solar cells, 2020](#)

2020: N.Y.). 1070 citations.

[Conformal quantum dot-SnO₂ layers as electron transporters for efficient perovskite solar cells, 2022](#)

2022: N.Y.). 1049 citations.

[Efficient perovskite solar cells via improved carrier management](#)

2021: Nature. 2342 citations.

[Controlled growth of perovskite layers with volatile alkylammonium chlorides](#)

2023: Nature. 1172 citations.

[Pseudo-halide anion engineering for \$\alpha\$ -FAPbI₃ perovskite solar cells](#)

2021: Nature. 2463 citations.

[Inactive \(PbI₂\)₂RbCl stabilizes perovskite films for efficient solar cells, 2022](#)

2022: N.Y.). 942 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 9640 out of 9751 articles. 2690 of these articles came from these sources.

Source	Articles (last five years)
Solar RRL	486
ACS applied materials & interfaces	426
ACS Applied Energy Materials	294
Advanced Functional Materials	293
Advanced Energy Materials	253
Chemical Engineering Journal	226
Advances in Materials	225
Solar Energy	168
Journal of Materials Chemistry A	161
Small (Weinheim an der Bergstrasse, Germany)	158

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 9732 of 9751 articles. 605 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Shengzhong Liu	135	42.85
Shengzhong (Frank) Liu	98	40.32
Yi-Bing Cheng	92	16.98
Kai Wang	90	25.28
Jihuai Wu	87	13.49
Liming Ding	87	29.68
Yi-Bing Cheng	86	24.94
Mohammad Khaja Nazeeruddin	79	32.38
Nam-Gyu Park	75	23.19
Nam-Gyu Park	75	61.81

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 9489 of 9751 articles. 2177 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Chinese Academy of Sciences - China	841	10.74
University of Chinese Academy of Sciences - China	301	9.23
Peking University - China	233	9.34
Soochow University - China	215	8.02
École Polytechnique Fédérale de Lausanne - Switzerland	212	16.26
Huazhong University of Science and Technology - China	208	11
Northwestern Polytechnical University - China	197	9.76
Wuhan University of Technology - China	197	7.35
Sungkyunkwan University - South Korea	183	12.32
Southern University of Science and Technology - China	168	11.17

Organization types

This table covers the 9288 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 9751 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
1.04%	40.19%	9.96%	3.04%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 444 out of 9751 articles. 174 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Applied Physical Sciences (United States)	32	27.46
Shanghai Fudan Microelectronics (China)	22	9.56
Solaronix (Switzerland)	19	3.96
Korea Electric Power Corporation (South Korea)	17	4.26
Zhejiang Energy Group (China)	17	5.18
Holst Centre (Netherlands)	15	3.32
Toyota Motor Corporation (Belgium)	14	7.21
Interface (United States)	13	4.25
Électricité de France (France)	13	2.17
Swiss Center for Electronics and Microtechnology (Switzerland)	12	7.33

Top funders

Data available for 8193 out of 9751 recent articles. 5388 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	3895	7.06
Ministry of Education - China	996	9.34
Ministry of Science and Technology - China	765	9.01
National Key Research & Development Program of China	601	9.59
National Research Foundation of Korea - South Korea	445	6.65
Ministry of Civil Affairs - China	369	8.44
Guangdong Province - China	291	8.05
National Research Foundation - South Korea	281	7.36
United States Department of Energy - United States	280	12.90
National Science Foundation - United States	268	7.79

Funder types

This table covers the 7161 articles in this cluster from the last five years with data about their funding organization types (out of 9751 articles total).

Commercial	Education	Nonprofit	Government
1.11%	12.13%	8.71%	78.06%

Patents and industry

1743 patents cite articles in this cluster (**99.72th** percentile among all clusters)

10.43% of articles in the cluster are cited by patents

1.02% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
METHODS FOR PRODUCING AND USING PEROVSKITE MATERIALS AND DEVICES THEREFROM	63
METHODS OF PREPARATION OF ORGANOMETALLIC HALIDE STRUCTURES	61
SOLAR CELL	60
METHODS FOR PRODUCING SINGLE CRYSTAL MIXED HALIDE PEROVSKITES	60
PEROVSKITE PHOTOVOLTAIC DEVICE	60
DOPED PEROVSKITE HAVING IMPROVED STABILITY, AND SOLAR CELLS MADE THEREOF	59
PEROVSKITE THIN FILMS HAVING LARGE CRYSTALLINE GRAINS	43
SEQUENTIAL PROCESSING WITH VAPOR TREATMENT OF THIN FILMS OF ORGANIC-INORGANIC PEROVSKITE MATERIALS	42
METHODS FOR FORMING A PEROVSKITE SOLAR CELL	42

Top patent assignees

Patent assignee	Same-cluster citations
牛津大学科技创新有限公司	91
NANYANG TECHNOLOGICAL UNIVERSITY	86
OSTEL MIEJI, MURAT	80
옥스포드 유니버시티 이노베이션 리미티드	79
ALLIANCE FOR SUSTAINABLE ENERGY LLC	262
UNIVERSITY OF NAT TAIWAN NORMAL	173
KING ABDULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY	148
OXEHEALTH LIMITED	145
ECOLE POLYTECHNIQUE FEDERALE DE LAUSANNE	106

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
467	2597	perovskite light-emitting diodes, external quantum efficiency, Halide Perovskite Nanocrystals, Metal halide perovskites, blue perovskite LEDs
7842	2587	perovskite solar cells, solar cells, lead-free perovskite solar, Tin halide perovskites, halide perovskite nanocrystals
27446	1846	perovskite solar cells, solar cells, inorganic perovskite solar, power conversion efficiency, PSCs
5064	1757	perovskite solar cells, solar cells, halide perovskites, metal halide, two-dimensional hybrid perovskites
13863	1694	perovskite X-ray detectors, perovskite single crystals, X-ray Imaging, Metal Halide Perovskites, X-ray detection
21952	1604	perovskite solar cells, solar cells, tandem solar, All-perovskite tandem solar, Solar cell efficiency
4729	1540	lead halide perovskites, perovskite solar cells, lead halide, Halide perovskite materials, solar cells
17940	1344	perovskite solar cells, solar cells, Stable perovskite solar, hole transport materials, HTMs
54348	1024	perovskite solar cells, solar cell, solar cell capacitance, based perovskite solar, hole transport layer
70725	219	Halide Perovskite, Perovskite Quantum Dots, efficient photocatalytic hydrogen, Metal Halide, charge separation

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
4729	1744	lead halide perovskites, perovskite solar cells, lead halide, Halide perovskite materials, solar cells
21952	1643	perovskite solar cells, solar cells, tandem solar, All-perovskite tandem solar, Solar cell efficiency
5064	1264	perovskite solar cells, solar cells, halide perovskites, metal halide, two-dimensional hybrid perovskites
27446	1030	perovskite solar cells, solar cells, inorganic perovskite solar, power conversion efficiency, PSCs
17940	888	perovskite solar cells, solar cells, Stable perovskite solar, hole transport materials, HTMs
7842	793	perovskite solar cells, solar cells, lead-free perovskite solar, Tin halide perovskites, halide perovskite nanocrystals
467	706	perovskite light-emitting diodes, external quantum efficiency, Halide Perovskite Nanocrystals, Metal halide perovskites, blue perovskite LEDs
13863	520	perovskite X-ray detectors, perovskite single crystals, X-ray Imaging, Metal Halide Perovskites, X-ray detection
4509	221	density functional theory, hydrogen evolution reaction, functional theory calculations, electronic structure, photocatalytic OER catalysts
3472	170	Organic Solar Cells, inverted organic solar, electron transport layer, solar cell devices, power conversion efficiency

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
barnesgroup/CL/Driftfusion	6	48
WMD-group/polytype	2	11
ej-jp/perovskino	1	0
WPT-Lab124/Modified-Diode-Model	1	3
EugenZimmermann/matlab-keithley2450	1	4
marcus-cmc/Shockley-Queisser-limit	1	48
i-MEET/boar	1	8
JohnMarohn/FreqDemod	1	4
SMTG-Bham/doped	1	132
PV-Lab/Autocharacterization-Bandgap	1	4

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).